Tool-Calling LLMs

- Developers write code and documentation, and prompt the model with a list of available tools

- LLMs can be trained to call developer-defined functions by predicting special tokens

- Models predict what tool to use and what inputs to give the tool for a given query

- Not generating any code itself!

- Provide the models with additional capabilities or improve on weaknesses

- e.g. calculator, internet search, web browser, query other LLMs, use real APIs, operate a computer

5

```python
def get_weather(self, location: str):
    """Get the weather in a city."""
    return weather_api.get(location)
```

<|im_start|>system
You can use the following functions:
get_weather(location: str)<|im_end|>

<|im_start|>assistant

<tool_call>

{"name":

"get_weather",

"params":

{

"location":

"Philadelphia,

PA''

}}

</tool_call><|im_end|>

```
>>> get_weather("Philadelphia, PA")
```

It's currently 75F and cloudy in Philadelphia, PA.

```
<|im_start|>assistant
<tool_call>
{"name": "get_weather", "params": {
  "location": "Philadelphia, PA"
}}
</tool_call><|im_end|>
```

Tool-Calling LLMs

- Developers write code and documentation, and prompt the model with a list of available tools
- LLMs can be trained to call developer-defined functions by predicting special tokens
- Models predict what tool to use and what inputs to give the tool for a given query
  - Not generating any code itself!
- Provide the models with additional capabilities or improve on weaknesses
  - e.g. calculator, internet search, web browser, query other LLMs, use real APIs, operate a computer

```
def get_weather(self, location: str):
    """Get the weather in a city."""
    return weather_api.get(location)

<|im_start|>system
You can use the following functions:
get_weather(location: str)<|im_end|>

<|im_start|>assistant
<tool_call>
{"name": "get_weather", "params": {
  "location": "Philadelphia, PA"
}}
</tool_call><|im_end|>

>>> get_weather("Philadelphia, PA")
It's currently 75F and cloudy in
Philadelphia, PA.
```

Tool-Calling LLM Agents

Just like they have back home in RL